

Pramaan — EPC Deviation Intelligence

Spec-to-Site Deviation Sentinel + Commissioning Risk Twin · ET AI Hackathon 2026 · Problem Statement 4

Repository: github.com/bansalbhunesh/parth · Live: parth-tan.vercel.app/judge · API: parth-3puc.onrender.com/health

Phase-2 deadline (live Unstop record): **2026-07-22, 23:59:21 IST**

The wedge: Pramaan converts a spec/submittal mismatch into a cited commissioning failure mode and the schedule window to act — *before* the failed test becomes a project delay. Not "RAG + agents + dashboard"; the trace from owner requirement → deviation → commissioning test → lead-time-to-failure, with evidence at each step.

Evidence labels used throughout this document: **MEASURED** observed in a run/test **SYNTHETIC** authored corpus, true by construction **DETERMINISTIC OFFLINE** reproducible with no API key **LIVE-MODEL** varies run-to-run **SCENARIO** modeled assumption, not validated.

1. The problem

In a hyperscale data-centre EPC build, the **design basis**, **vendor submittals**, and **governing standards** live in different systems and are reviewed by different people. A vendor deviation missed at submittal review resurfaces months later at commissioning — as a failed acceptance test, rework, and schedule slip. Found on submittal day it is a one-line RFI; found at commissioning it is a schedule event.

Market context **SCENARIO**: India data-centre capex is large and growing (IBEF, secondary research: ~US\$10B in 2025 toward ~US\$22B by 2030; JLL primary release is more conservative — 791 MW added by 2026, US\$5.7B investment). These frame the opportunity; they are not Pramaan-specific results.

2. What Pramaan does

- Reconciles a design specification against a vendor submittal and the governing standards, and reports each genuine deviation with the standard it violates.
- For each deviation, predicts the **commissioning test** that will fail and the **lead-time-in-weeks** to fix it if caught now — the commissioning-risk twin.
- Renders an evidence chain (required value, submitted value, source citation, confidence) and exports an audit trail for the commissioning authority.
- Degrades gracefully: when the LLM is unavailable, an independent rule-based detector still catches the headline numeric shortfalls — no silent zeros, and never the answer key (see §5).

3. Architecture (as actually executed)

Pramaan is **one compliance/Cx reasoning graph plus three connected deterministic intelligence services** — not five co-operating AI agents. The runtime compliance graph is:

`ingest → load_standards → validate → reconcile → retrieve → critique → cx_predict → format_output`

- `validate` is a routing gate; `reconcile / retrieve / critique` form a bounded retrieval + reflexion loop; `cx_predict` maps each deviation to its commissioning test via a rule table (no per-deviation LLM call).
- The standalone **extraction** module is *not* in the executed graph, and the **RFI Copilot** is a separate API workflow — stated plainly rather than diagrammed as active agents.
- The three deterministic services — **schedule risk** (graph CPM + seeded beta-PERT Monte Carlo), **supply chain** (closed-form ETA/risk over static data), and the **project/Cx graph** — are separate endpoints, not agents in the compliance graph.

4. Deployment & live behaviour

LIVE DEMO MEASURED

Judge page + API return 200; a live Gemini text comparison and image (vision) analysis both returned valid findings during an external audit.

LATENCY MEASURED

Model calls run ~13–33 s; the UI streams reasoning and caps the wait, degrading to the rule engine on timeout.

OCR MEASURED

OCR runs only where Tesseract is installed (the `Dockerfile.backend` image). The default Render/Compose deploy does not bundle it and returns a clear "OCR unavailable" message.

5. Scientific integrity of the analysis path

User-facing analysis **never substitutes the answer key**. When the LLM pipeline returns empty (throttled / no key), the corpus endpoints degrade to the independent rule detector over the same spec + submittal, labelled `analysis_mode:"rule"`, or report `analysis_mode:"unavailable"` — never the seeded ground-truth labels. The seeded fixture is reachable only via an explicit `?seeded_demo=true` opt-in, stamped `analysis_mode:"seeded_demo"` with a disclaimer. The copilot's offline fallback is labelled `mode:"offline-fallback"`. This is verified by automated tests that assert seeded identifiers never leak into the default path.

6. Evaluation methodology & results

6a. Synthetic portfolio — breadth check SYNTHETIC

A 12-project, 11-country synthetic corpus with **seeded** deviations gives repeatable breadth coverage. Its P/R/F1 read **1.000 by construction** — it exercises the plumbing across formats and standards; it is *not* evidence of external accuracy, and we label it that way rather than headline it.

6b. Team-authored evaluation pairs — the credibility signal

Fifteen team-authored pairs whose values are **cited from public product datasheets and standards** (Vertiv, Cummins, STULZ, ABB, Tate ConCore, Schneider Canalis, Raritan, Xtralis VESDA, Distech, and others). They are realistic engineering documents, *not* downloaded immutable vendor files. Across them are **27 hard deviation claims + 1 deliberately contested case**.

Layer	What it proves	Result
Deterministic offline DETERMINISTIC OFFLINE	Rule detector recovers headline numeric shortfalls with no API key; clears compliant/true-negative values	4 of the 27 hard claims + true-negatives; 0 false positives , reproducible (<code>python eval/real_pairs_offline.py</code>)
Live-model LIVE-MODEL	Frontier model recovers the reasoning-dependent remainder (derived arithmetic, recalled GWPs, categorical/omission findings)	The other 23 hard claims require a live model; recall/precision vary run-to-run and are reported per dated run, not as a fixed score

Honest scoring. The live scorer now performs **one-to-one (bipartite) matching** — each finding is credited to at most one ground-truth label and vice-versa, so a single broad finding cannot inflate recall, and unmatched findings are counted as false positives (never a negative "extra"). It reports **two** recall numbers: an *executed-only* recall and a *primary* recall that counts not-run (quota/outage) pairs as misses. One case (a 30°C supply-air setpoint, within ASHRAE A1 allowable but above recommended) is deliberately contested and scored against us, so the honest precision sits near ~0.9, not a suspicious 1.000.

Independence caveat. On one BMS pair, a live run surfaced two consequences we had not listed and we adopted them into ground truth; labels added after seeing the model's output are no longer independent of it, and we say so. A fully independent, frozen, primary-source benchmark with pre-registered labels is the next validation milestone.

6c. Example deviations (illustrative, from the team-authored pairs)

Parameter	Required (standard)	Submitted (cited value)	Layer
UPS battery autonomy	≥ 10 min (Uptime Tier IV)	7 min (Vertiv GXT5-1000)	OFFLINE
UPS online efficiency	≥ 96%	95.9% online (99% is ECO)	OFFLINE
Generator on-site fuel	≥ 48 h	38.8 h (derived: 4,000 gal ÷ 103 GPH)	LIVE-MODEL

Generator emissions	EPA Tier 4	EPA Tier 2 (Cummins QSK60)	LIVE-MODEL
Refrigerant GWP	≤ 750 (EU F-Gas)	R-410A GWP 2,088 (model-recalled)	LIVE-MODEL
Raised-floor point load	≥ 1,500 lbf	1,250 lbf (Tate ConCore 1250 ceiling)	OFFLINE
Busway short-time withstand	≥ 65 kA/1 s	50 kA/1 s (Schneider Canalis KTA10)	OFFLINE
Supply-air setpoint (contested)	≤ 27 °C recommended	30 °C (within A1 allowable)	CONTESTED

Compliant values (e.g. IP54 ≥ IP42, 415 V, 10 s NFPA start) are deliberately present and are *not* flagged — a precision / true-negative check.

6d. Reproduce it

- `python -m pytest tests -q` — 480+ tests, no API key **MEASURED**
- `python eval/real_pairs_offline.py` — deterministic offline recall + 0 false positives, exits 0
DETERMINISTIC OFFLINE
- `make eval-real` — the live-model layer; prints executed-only *and* primary recall with per-pair TP/FP/FN
LIVE-MODEL
- `python scripts/verify_live.py` — end-to-end checks against the deployment **LIVE-MODEL**

7. Business impact

The asymmetry is real: an undetected deviation can become a multi-week commissioning slip, while an analysis costs cents. But the **pricing, ROI multiple, payback, and margin figures are modeled hypotheses** **SCENARIO** — not validated by a paid pilot, LOI, or customer, and the ROI calculator is a sensitivity model, not realized savings. We have **not yet** run a timed manual-coordination study, so we make no measured hours-saved claim. The defensible business statement today: "The cost of late nonconformance can be high; Pramaan is designed to move detection to submittal review."

8. Engineering quality

TESTS **MEASURED**

480+ automated tests (481 locally); backend coverage ~81%. Full CI: Python tests + lint, frontend types/tests/build, Docker.

RESILIENCE
DETERMINISTIC OFFLINE

No API key, malformed PDFs, cold backend all return 200; `/11m-check` surfaces the true LLM status.

REPRODUCIBILITY
DETERMINISTIC OFFLINE

Deterministic evals and unit tests run offline; live-model results are labelled and vary run-to-run.

Prototype, not a hardened production service. There is no authentication, rate limiting, tenant isolation, or upload/dependency hardening yet; synchronous LLM calls and single-instance free-tier hosting are not a production-load story. These are named limitations, not claims.

9. Security, standards & IP posture

- No API keys are committed; credentials are external secrets. Path-traversal, malformed-input, output-escaping, and upload-size tests exist; error bodies never leak stack traces.
- Third-party standards content is a team-authored paraphrase from public/secondary references — proprietary standard text is not redistributed; product/standard names are trademarks of their owners and Pramaan is not endorsed by them.

- Known follow-ups (P1): auth + rate limits on public LLM endpoints, upload/dependency hardening, and a `THIRD_PARTY_NOTICES.md` with per-artifact source/version/hash.

10. Honest limitations & roadmap

- **Schedule risk is simulation**, not field-calibrated prediction; deviation probabilities and "actual delays" are seeded, not empirical.
- **Supply-chain intelligence is illustrative** — closed-form over static data, no live shipment/route feed yet.
- **No measured coordination-hours study** yet; the 45-day figure was a UI constant and is not presented as a result.
- **Next milestones:** a frozen independent benchmark (primary-source PDFs, hashes, pre-registered labels, clean negatives, one-to-one TP/FP/FN with variance); a timed human baseline; auth + rate limiting; and either shipping OCR in the hosted image or dropping OCR claims.

11. Rubric mapping (PS4 weights: 25 / 25 / 20 / 15 / 15)

Criterion	How Pramaan addresses it	Honest status
Innovation (25%)	Spec → deviation → commissioning test → lead-time chain; a distinctive wedge beyond submittal review	Strong — the moat is the Cx-test / lead-time link, not module count
Business impact (25%)	Cost-of-late-nonconformance thesis; scenario ROI model	SCENARIO no measured hours-saved study yet
Technical excellence (20%)	Real LangGraph + LLM, schedule simulation, 480+ tests, CI, live LLM/vision	MEASURED answer-key fallback removed; scorer corrected
Scalability (15%)	Modular, containerized; deterministic services run at ~142 systems/s	Prototype: no auth / persistence / queue; throughput is rule execution, not LLM
User experience (15%)	Focused Judge Mode + streaming reasoning; polished desktop UI	Known mobile / a11y / latency follow-ups (P1)

12. Anticipated judge questions (honest answers)

Is the data real? The 12-project portfolio is synthetic regression data. The evaluation pairs are team-authored with values cited from public product figures; 4 claims are checked deterministically offline, 23 require a live model. A fully independent, frozen benchmark is the next milestone.

Is schedule risk predictive? Today it is a seeded CPM / Monte-Carlo scenario engine, not calibrated on field actuals. That calibration is the next validation step.

Why "agents"? The LLM is used only for cross-document semantic reconciliation; validation, Cx mapping, schedule propagation, and guardrails are deterministic so outputs stay inspectable. It is one reasoning graph plus three deterministic services, not five agents.

What did users save? We make no measured field-savings claim yet; a timed human baseline is planned.

Evidence-safe proof points: "480+ automated tests pass; backend coverage ~81%." · "The public deployment performed a live Gemini text and vision analysis during an external audit." · "The schedule service runs seeded CPM/Monte-Carlo simulation; delay results are simulated, not field predictions." · "The 12-project portfolio is synthetic regression coverage." · Every metric above carries an evidence label. Results are observations at the audited commit, not evergreen badges.